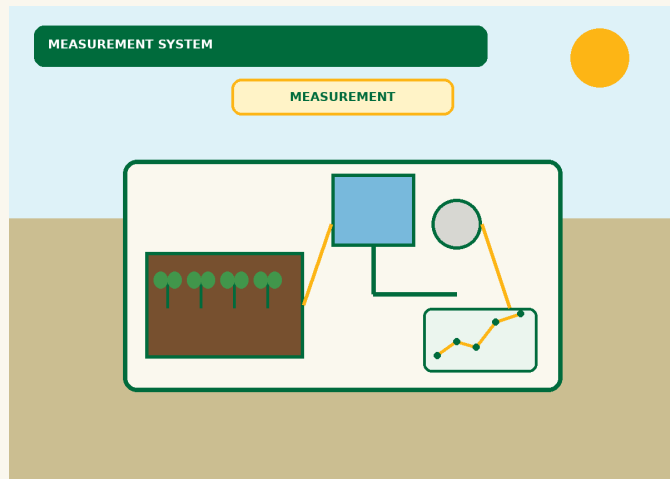




MEASUREMENT

MEASUREMENT SYSTEM

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A measurement system includes the measurand, sensor, signal conditioning, data acquisition, calibration, metadata, and interpretation.

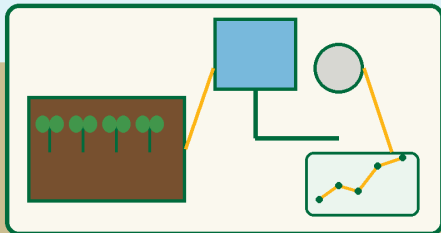
ANALYSIS / DESIGN TASK

Map the measurement chain for one variable.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

SENSOR SPECIFICATION

SPECIFICATION



SPECIFICATION

SENSOR SPECIFICATION

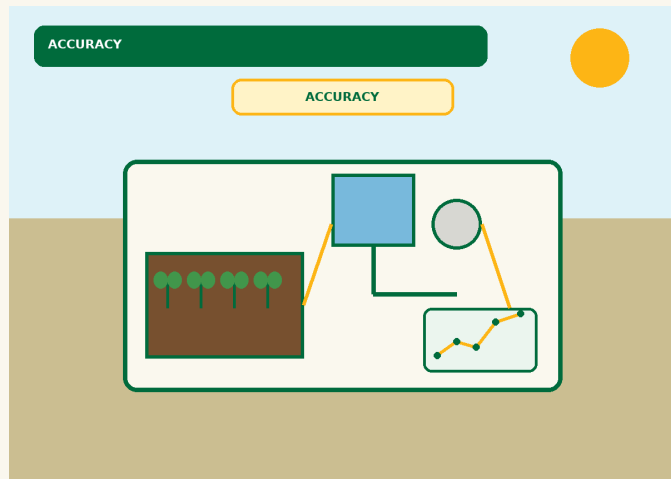
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Sensor specifications define range, accuracy, resolution, response, environmental limits, power, and interface requirements.

ANALYSIS / DESIGN TASK

Evaluate whether a sensor fits the intended use.



ACCURACY

ACCURACY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

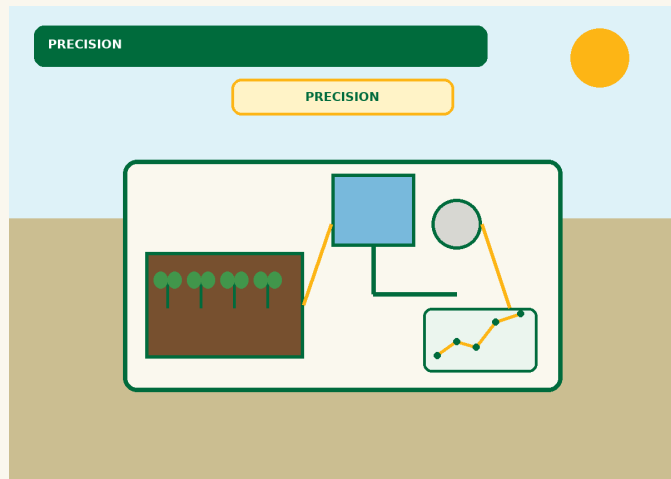
TECHNICAL CONCEPT

Accuracy describes closeness to a reference or true value under specified conditions and should not be confused with precision.

ANALYSIS / DESIGN TASK

Calculate measurement error against a reference.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



PRECISION

PRECISION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

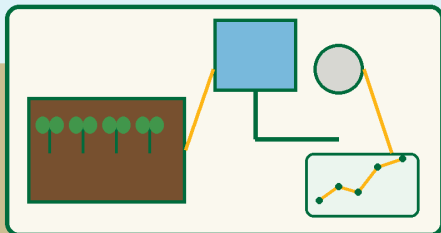
Precision describes agreement among repeated measurements and can be quantified using spread statistics.

ANALYSIS / DESIGN TASK

Calculate standard deviation of repeated readings.

BIAS ERROR

BIAS



BIAS

BIAS ERROR

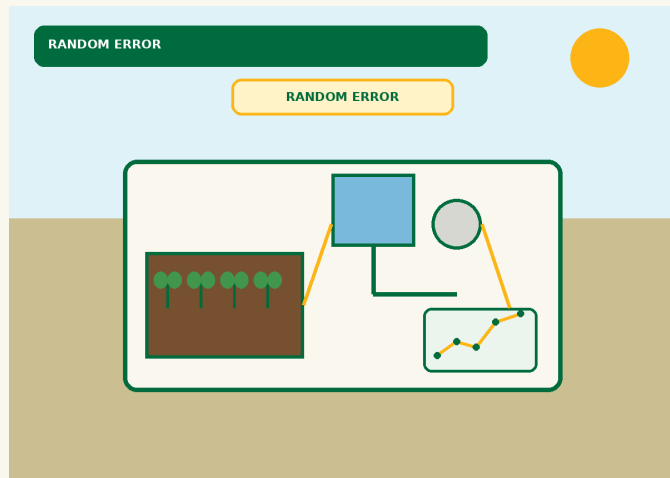
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Bias is a systematic measurement error that shifts readings consistently away from the target value.

ANALYSIS / DESIGN TASK

Estimate bias from calibration data.



RANDOM ERROR

RANDOM ERROR

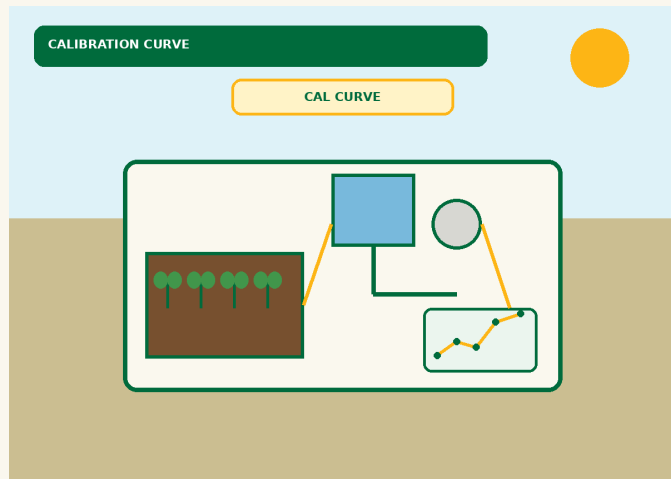
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Random error produces unpredictable measurement variation and contributes to uncertainty in repeated observations.

ANALYSIS / DESIGN TASK

Identify random variation in repeated measurements.



CAL CURVE

CALIBRATION CURVE

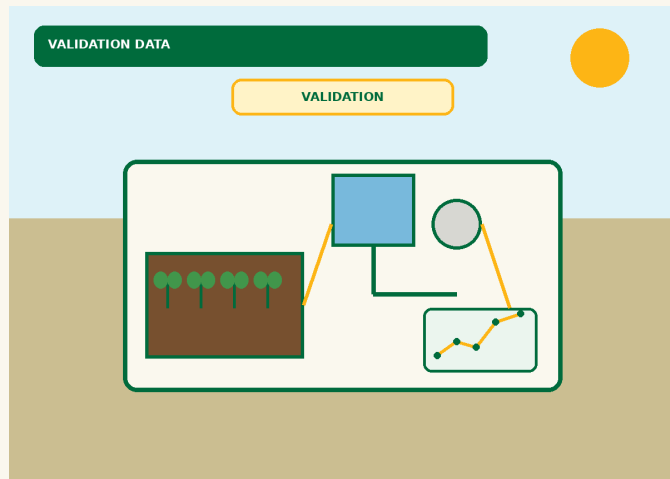
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A calibration curve relates sensor output to known reference values and may be used to convert raw signals into engineering units.

ANALYSIS / DESIGN TASK

Fit or interpret a simple linear calibration.



VALIDATION

VALIDATION DATA

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

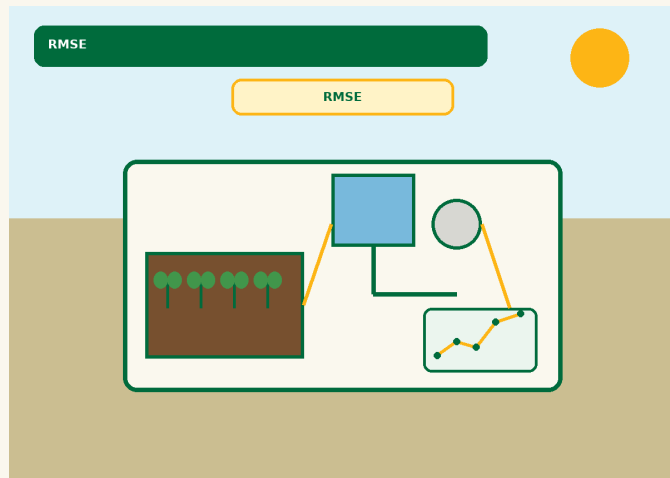
TECHNICAL CONCEPT

Validation uses independent reference observations to test sensor performance after calibration or under field conditions.

ANALYSIS / DESIGN TASK

Compare predicted and reference values.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



RMSE

RMSE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

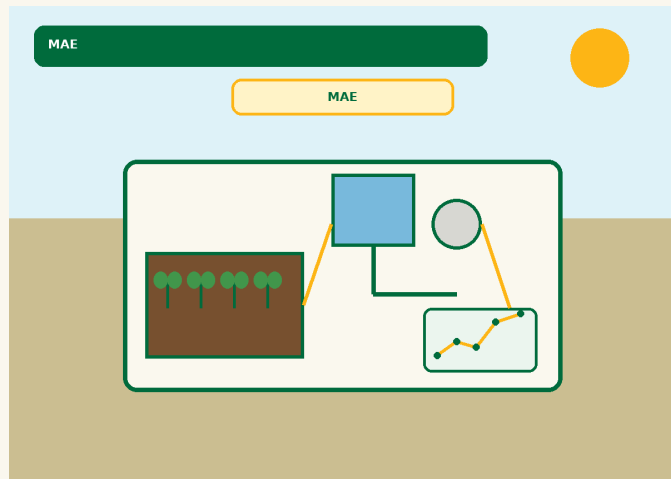
TECHNICAL CONCEPT

Root mean square error summarizes the magnitude of prediction or measurement errors while giving larger errors greater weight.

ANALYSIS / DESIGN TASK

Calculate RMSE from a small dataset.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



MAE

MAE

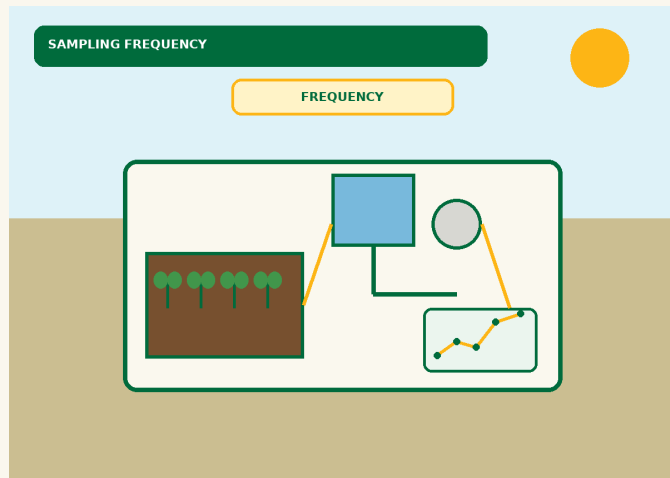
TECHNICAL CONCEPT

Mean absolute error is the average absolute difference between measured or predicted values and references.

ANALYSIS / DESIGN TASK

Calculate MAE and compare with RMSE.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



FREQUENCY

SAMPLING FREQUENCY

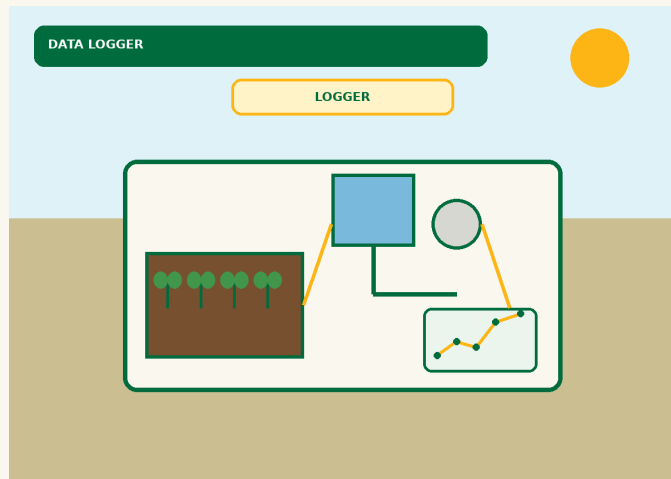
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Sampling frequency is how often a signal is measured and must be appropriate for the rate at which the process changes.

ANALYSIS / DESIGN TASK

Choose intervals for fast and slow variables.



LOGGER

DATA LOGGER

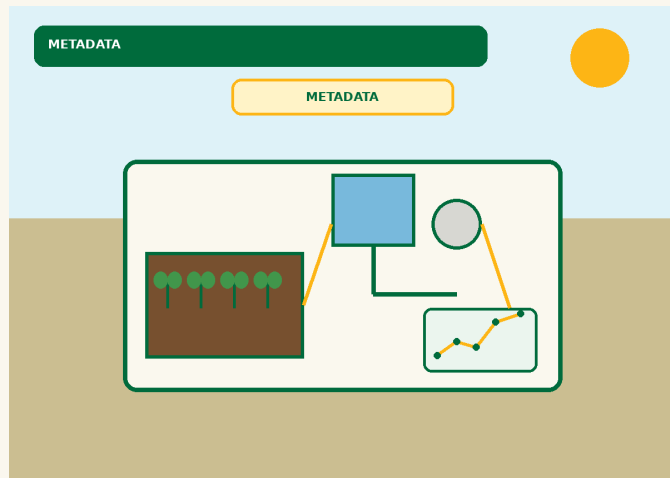
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A data logger time-stamps, stores, and sometimes transmits measurements from one or more sensors.

ANALYSIS / DESIGN TASK

Audit logger channels and timestamps.



METADATA

METADATA

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

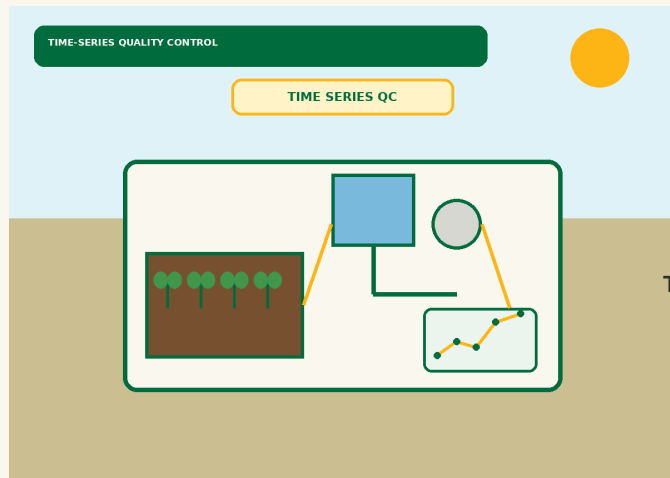
TECHNICAL CONCEPT

Metadata document sensor identity, units, location, depth, calibration, timestamps, maintenance, and processing history.

ANALYSIS / DESIGN TASK

Create a metadata record.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



TIME SERIES QC

TIME-SERIES QUALITY CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

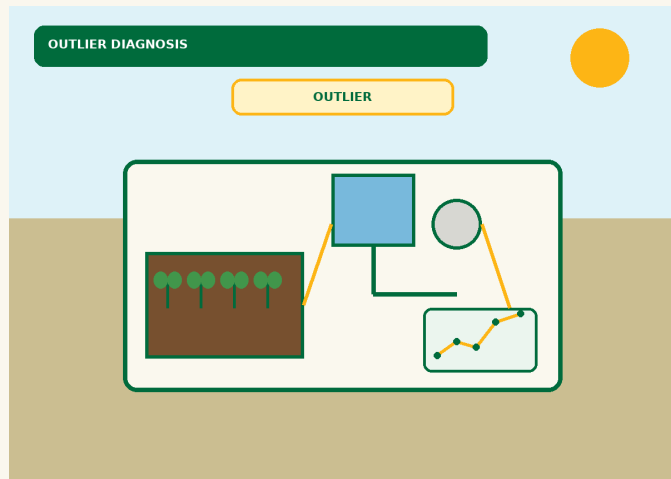
TECHNICAL CONCEPT

Time-series quality control detects missing values, impossible values, spikes, drift, timestamp errors, and sensor failures.

ANALYSIS / DESIGN TASK

Flag anomalies using defined rules.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



OUTLIER

OUTLIER DIAGNOSIS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

An outlier should be investigated using context, neighboring measurements, instrument status, and independent evidence before exclusion.

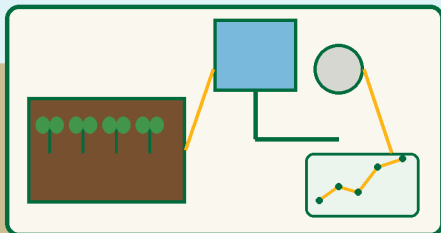
ANALYSIS / DESIGN TASK

Diagnose a suspicious observation.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

MISSING-DATA PATTERN

MISSING DATA



MISSING DATA

MISSING-DATA PATTERN

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

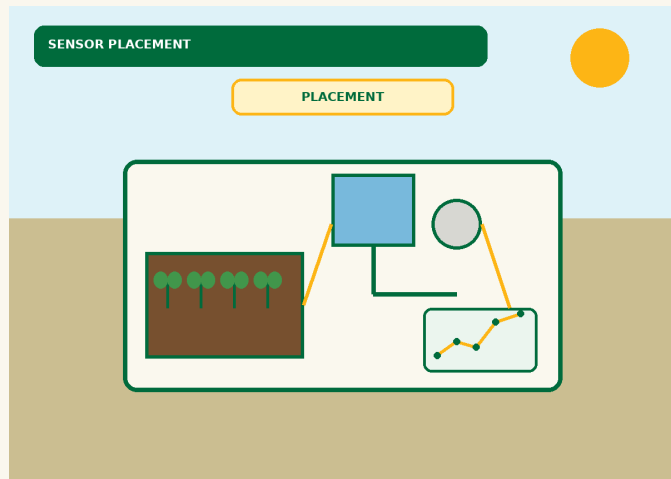
TECHNICAL CONCEPT

Missing-data patterns can reveal power, communication, logger, sensor, or maintenance problems and affect analysis.

ANALYSIS / DESIGN TASK

Classify gaps in a dataset.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



PLACEMENT

SENSOR PLACEMENT

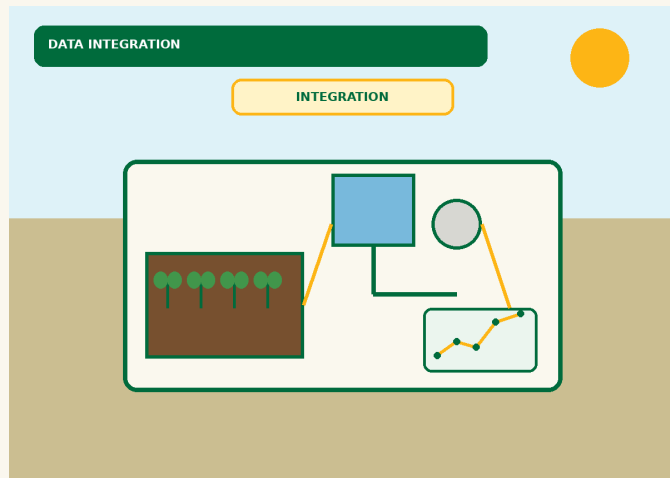
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Placement determines whether measurements represent the intended spatial location, depth, exposure, and process.

ANALYSIS / DESIGN TASK

Evaluate two possible sensor locations.



INTEGRATION

DATA INTEGRATION

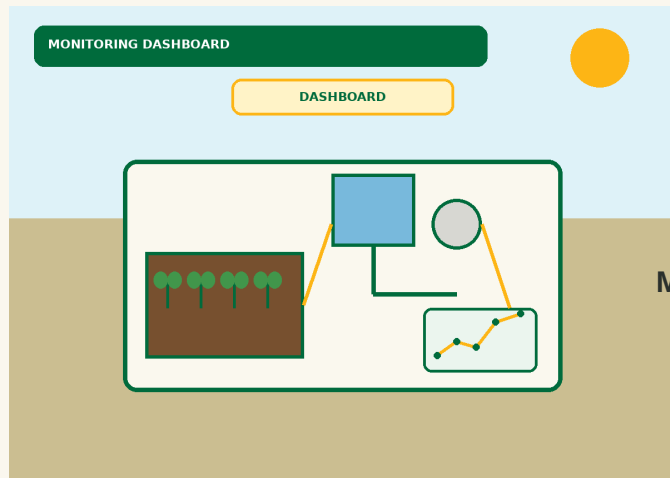
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Data integration combines measurements from multiple variables, sensors, locations, or time scales into a consistent analytical dataset.

ANALYSIS / DESIGN TASK

Align two time series by timestamp.



DASHBOARD

MONITORING DASHBOARD

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

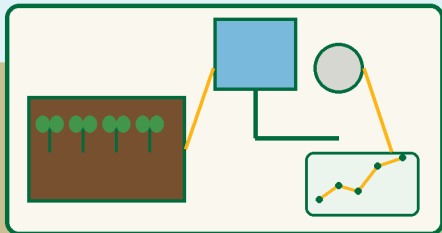
A monitoring dashboard summarizes current state, trends, thresholds, data quality, and operational indicators for decision support.

ANALYSIS / DESIGN TASK

Design a minimal dashboard layout.

MONITORING SYSTEM EVALUATION

EVALUATE



EVALUATE

MONITORING SYSTEM EVALUATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Evaluation considers sensor fitness, calibration, uncertainty, placement, data continuity, integration, maintenance, and decision value.

ANALYSIS / DESIGN TASK

Audit the monitoring system and prioritize improvements.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE